Journal of Nuclear Materials 206 (1993) 200–211
North-Holland

journal of nuclear materials

Impact of glissile interstitial loop production in cascades on defect accumulation in the transient

H. Trinkaus [a], B.N. Singh [b] and A.J.E. Foreman [c]

[a] *Institut für Festkörperforschung, Forschungszentrum Jülich, Postfach 1913, D-52425 Jülich, Germany*
[b] *Materials Department, Risø National Laboratory, DK-4000 Roskilde, Denmark*
[c] *Radiation Damage Department, Reactor Services, Harwell Laboratory, Oxfordshire OX11 0RA, United Kingdom*

Defect interactions in displacement cascades produced during high energy particle irradiation result in the annihilation and clustering of vacancies and self-interstitial atoms (SIAs) within the cascade volume. SIA clusters produced in the form of small glissile loops may glide and reach extended sinks such as dislocations and grain boundaries. In the present paper, this process is suggested to represent an efficient driving force for void swelling, in particular in regions adjacent to grain boundaries. For describing the evolution of the point defect and defect cluster concentrations in the initial stage of irradiation a set of kinetic equations taking the escape of one-dimensionally gliding SIA loops into account is proposed. The calculation of the vacancy supersaturation as a function of dose shows that at sufficiently high dislocation densities, microvoids that are not stable in the quasi-steady-state at higher doses could be stabilized during a transient maximum of the vacancy supersaturation at lower doses. The enhanced swelling observed adjacent to grain boundaries is attributed to the escape of SIA loops to these planar sinks. The width of the swelling zone is shown to be of the order of the mean range of one-dimensionally gliding loops. From the maximum swelling rate, the fraction of SIAs produced in the form of glissile loops is estimated.

1. Introduction

The recoil energy transferred from a projectile particle to a lattice atom plays a crucial role in defect production. Low recoil energies transferred by electrons, for instance, result in the formation of single vacancies and self-interstitial atoms (SIAs) whereas high recoil energies transferred by energetic neutrons and ions induce multidisplacement cascades. In the latter case, both vacancies and SIAs are produced in a highly localized and segregated fashion. At the end of the collisional phase a vacancy rich core is surrounded by a high concentration of SIAs at the periphery.

The primary cascade damage has been investigated by experimental methods as well as by computer simulations. Transmission electron microscopy (TEM), for instance, has shown that the vacancy rich cascade core tends to collapse [1–3] to form a dislocation loop or a stacking fault tetrahedron or, less frequently, a three-dimensional voidlike vacancy aggregate. Diffuse X-ray scattering on fast neutron irradiated metals at temperatures below stage I provide evidence for spontaneous SIA cluster formation in cascades [4–6]. Recent molecular dynamics (MD) studies [7,8] demonstrate that efficient SIA clustering occurs already during the cooling down phase of the cascade.

Small clusters consisting of two or three vacancies [9] or SIAs [10] are known to be mobile, similar to or even more mobile than the corresponding monodefect. Larger clusters tend to form dislocation loops. The possible migration modes of such defect clusters are climb and glide. The latter process is, however, much faster than the former. Since small loops are commonly assumed to be faulted their glide motion is considered to require the removal of the stacking fault by partials driven by internal stress fields. The evolution of SIA clusters in annealing stage II after low temperature electron [11,12] and fast neutron irradiation [12–14], for instance, may be explained as a result of a combined glide and climb motion of SIA clusters under the action of strong elastic interaction forces. Loop glide has been directly observed in TEM [15,16]. Recent MD studies [8] have shown that small loops consisting of 4, 5 and 6 SIAs prefer to be unfaulted and highly glissile.

Indirect evidence for the long-ranging one-dimensional SIA transport is provided by the observation of highly heterogeneous microstructures evolving in well annealed metals during low dose irradiations: SIAs and vacancies precipitate in different regions separated by several μm [17]; in virtually dislocation free regions, the SIAs corresponding to the voids are missing according to TEM observations [17,18] as well as lattice parameter measurements [19]; grain boundaries induce enhanced swelling in region adjacent to them extending over several μm [20–28]. These features cannot be understood in terms of a three- (or two-) dimensional diffusion reaction kinetics as will be shown in the present paper.

For the microstructural evolution in the temperature range of void swelling (between one third and one half of the melting temperature, T_m) it is important that SIA clusters are thermally relatively stable whereas vacancy clusters decay quickly under such conditions. This asymmetry concerning the form in which vacancies and SIAs survive intracascade recombination has been recently introduced as a "production bias" [29–33] which would become a potent driving force for void swelling if a significant fraction of the stable SIA clusters produced in cascades were annihilated preferentially at extended sinks such as dislocations, grain boundaries or surfaces. One of the possible mechanisms for cluster removal is dislocation sweeping which is discussed elsewhere in this issue [33]. In the present paper, the one-dimensional glide of SIA loops to extended sinks will be shown to represent a particularly efficient SIA cluster removal mechanism.

A brief discussion of possible effects of SIA loop glide has been presented recently [34]. In the following, some important aspects will be treated in greater detail. In section 2, the configurational stability and mobility of small SIA loops will be discussed. In section 3, reactions of glissile loops with other defects will be analysed. Section 4 is devoted to the effects of loop escape by glide on the early stages of the microstructural evolution during irradiation. In section 5, the enhanced swelling adjacent to grain boundaries will be analysed in terms of loop escape to such sinks.

2. Configurational stability and mobility of small SIA loops

Dislocation loops of SIA-type produced in cascades contain a few to a few tens of SIAs. This size range is at the upper boundary accessible to computer simulation and below the range where the elastic continuum can be applied reliably. Therefore, estimates of energies controlling stability and mobility of such defect clusters are necessarily uncertain.

Since small loops are commonly assumed to be faulted their glide motion is considered to require the removal of their stacking fault by thermal activation or by the action of internal stress fields. Recently, the one-dimensional random glide motion of a planar SIA cluster consisting of four SIAs has been observed in a MD study [8]. It has been shown in this study that small loops consisting of four, five and six atoms transform spontaneously from the faulted into the highly glissile unfaulted configuration. Present MD studies do not provide clear evidence whether larger loops produced in cascades are also glissile but the experimental studies quoted in the introduction give some indication that this assumption is realistic.

Another important aspect is the thermally activated change between different loop configuration. So far, it is not clear whether a barrier exists between a faulted and the corresponding unfaulted configuration (faulted $\rightleftarrows$ "mixed" $\rightleftarrows$ unfaulted) or whether a configuration of one kind always represents a barrier for a transition between two equivalent configurations of the other kind (for instance, unfaulted with a certain Burgers vector $\rightleftarrows$ faulted $\rightleftarrows$ unfaulted with an equivalent other Burgers vector). The above mentioned MD simulations of cascade defects in "Cu" provide some indication that larger loops (~ 15 SIAs) may be glissile at elevated temperatures which would be in favour of an activated transition from a stable sessile faulted to a metastable glissile unfaulted configuration. On the other hand, the change in the Burgers vector of a cluster consisting of four SIAs has been observed to occur in these studies. From the lifetime of one configuration a value of 0.4 eV is estimated for the barrier against this transformation. The height of this barrier is expected to increase strongly with increasing loop size. For large loops, a dislocation with a Burgers vector equal to the difference between the Burgers vectors of the two configurations would have to pass across the loop area. By the formation of kink pairs, an unfaulted loop could also change into another habit plane.

In pure metals, the mobility of glissile dislocation loop is certainly highly anisotropic since glide is much faster there than climb. Climb of small loops is probably controlled by dislocation core diffusion with an activation energy comparable with that for grain boundary diffusion (and would in this case be "conservative"). Activation energies for the glide of small loops may be estimated on the basis of the Peierls energy per dislocation core atom. Peierls stresses below 10^{-3} of the shear modulus μ, as commonly assumed for pure metals, yield estimates below 0.1 eV for loops containing less than 100 SIAs. In an undisturbed pure metal at temperatures of interest, a loop with such a low barrier against glide would perform an extremely fast random one-dimensional glide motion in the direction of its Burgers vector. Impurities are expected to reduce the mobility of glissile loops, possibly increasing their effective migration energy by a few tenths of an eV. Thermally activated changes of the Burgers vector in crystallographically equivalent directions would result in a zigzag glide motion made up of straight portions. The length of these straight portions is expected to increase strongly with increasing loop size according to the increase of the activation energy for such Burgers vector changes.

We conclude this section with the remark that, to our knowledge, no evidence for the glide of small vacancy loops exists so far. This asymmetry in the mobility of vacancy and SIA loops is probably due to the difference in the sign of the associated strain fields. A small vacancy loop is probably pinned by the poorly relaxed vacancy platelet located in the loop plane. Vacancy loops as well as SIA loops may, however, migrate by "conservative climb".

3. Reactions of glissile loops with other defects

A glissile loop will glide through undisturbed parts of the crystal lattice until it is trapped by another immobile or mobile defect. A detailed description of the corresponding reaction kinetics [35] is beyond the scope of the present paper. For the understanding of the simple approach used in the following, a few remarks on some key features are, however, helpful.

In the case of one-dimensionally diffusing defects, such as glissile loops of conserved Burgers vector, the probability of a defect to be trapped is quadratic in the trap density, i.e. very low for low trap densities, corresponding to a reaction kinetics of third order [36]. This is in contrast to the case of three-dimensionally diffusing defects for which the trapping probability is linear in the trap density corresponding to a reaction kinetics of second order. When the defect changes its straight diffusion path before it is trapped the trapping probability increases with decreasing straight diffusion range and approaches the value for three-dimensional diffusion when the straight diffusion range becomes smaller than the linear extension of the traps. The reaction kinetics of defects diffusing one-dimensionally on crystallographically equivalent but nonparallel paths corresponds to the reaction kinetics of two-dimensionally diffusing defects and is, like the latter that, of second order [37].

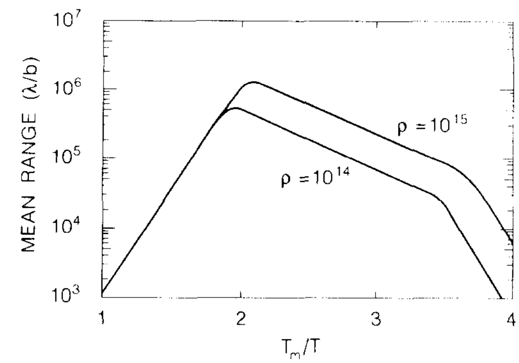

Fig. 1. Mean range normalized to the Burgers vector, λ / b, of a SIA loop gliding along a vacancy-containing but otherwise defect-free cylinder versus reciprocal homologous temperature, T_m / T, for two dislocation densities ρ.

The main effect of the long-ranging 1D SIA transport in glissile loops may be illustrated by considering the mean range of such loops in a crystal lattice containing other defects. We first consider the reaction of glissile SIA loops with single SIAs and vacancies. In the temperature range of void swelling which is of interest here, the absorption of single SIAs by glissile SIA loops may be neglected since the concentrations of both defects are very low because of their high mobility. The absorption of less mobile single vacancies by glissile SIA loops is, however, much more important and deserves some attention here. A SIA loop will disappear when it has swept as many vacancies as SIAs were initially contained in it. Let us assume that the lifetime of the loop is determined by this vacancy sweeping process. In this case, two limiting conditions may be distinguished: short lifetimes corresponding to short ranges for which refilling of swept parts of the glide cylinder with vacancies may be neglected, and long lifetimes corresponding to long ranges for which the vacancy concentration in the glide cylinder may be assumed to be equal to that in the environment. Which of these cases applies depends on temperature.

In fig. 1, the mean range normalized to the Burgers vector, λ/b, of a SIA loop gliding along a vacancy-containing but otherwise defect-free cylinder is plotted versus the reciprocal homologous temperature, T_m/T, for two dislocation densities, ρ, assumed to control the vacancy concentration below 0.5 T_m (the parameters used are given in table 1). Three characteristic ranges may be distinguished: a low temperature range where the refilling of swept parts of the glide cylinder with vacancies may be neglected, and medium and high temperature ranges where the vacancy concentration in the glide cylinder may be assumed to be equal to the

Table 1
Parameters used in the calculations

NRT displacement rate,	10^{-6} dpa/s
Effective displacement rate, P	10^{-7} dpa/s
Fraction of SIAs deposited in sessile clusters, ϵ_i^s	0.45
Fraction of SIAs deposited in glissile clusters, ϵ_i^g	0.05
Fraction of vacancies deposited in sessile clusters, $\epsilon_v = \epsilon_v^s$	0.5
Average number of defects in sessile clusters produced in cascades, n_i^s, n_v	25
Average number of SIAs in glissile loops	5
Recombination and trapping coefficients, a, a_{ni}, a_{nv}	5×10^{20} m^{-2}
Atomic volume, Ω	12×10^{-30} m^3
Shear modulus, μ	$35\ kT_m/\Omega$
SIA diffusion coefficient, D_i	
preexponential, D_{i0}	10^{-5} m^2 s^{-1}
migration energy, E_i^m	kT_m
Vacancy diffusion coefficient, D_v	
preexponential, D_{v0}	10^{-5} m^2 s^{-1}
migration energy, E_v^m	$7\ kT_m$
Self-diffusion coefficient, D_v^{sd}	
preexponential, D_0^{sd}	10^{-5} m^2 s^{-1}
activation energy, E^{sd}	$18\ kT_m$
Vacancy supersaturation in the presence of vacancy loops	exp[5 T_m/T]

mean radiation-induced and the thermal equilibrium vacancy concentration, respectively. Fig. 1 shows that vacancy sweeping allows an extremely wide spatial range for gliding SIA loops extending to several tens of μm for temperatures between one third and one half of T_m.

Next, we consider the trapping of glissile SIA loops by defect clusters and network dislocations. In the case of a crystal lattice distorted by such defects, it is useful to distinguish regions between the defects, where changes in the elastic interaction energy of glissile loop with the microstructure are smaller than the thermal energy kT, from the regions of strong elastic interaction around the defects. In the former region a glissile loop performs a random one-dimensional glide motion whereas in the latter it is either repelled or attracted by a defect. The volume fraction where random glide is possible increases with increasing temperature and decreases with increasing size of the glissile loop and with increasing defect density.

The trapping of one-dimensionally migrating defects by defect clusters such as cavities and small sessile dislocation loops may be characterized by the two types of parameters used in the collision theory: the effective trapping cross sections and the number densities of the trapping defects [38]. For the trapping by line defects such as dislocations, the effective trapping cross section reduces to an effective trapping diameter. For a defect of configuration i migrating one-dimensionally in a crystal containing a number density C_j of defects of configuration j with effective interaction cross section σ_{ij} and, in addition, a dislocation density ρ with an effective interaction diameter d_i, the reciprocal mean free path, $\kappa_i = \lambda_i^{-1}$, may be written as

$$\kappa_i = \lambda_1^{-1} = \sum_1 \sigma_{ij} C_j + d_i \hat{\rho}, \tag{1}$$

where $\hat{\rho} = \pi\rho/4$ is the dislocation line length projected on a plane perpendicular to the migration direction per unit volume. For a random distribution of the diffusion directions over equivalent directions in a cubic lattice the mean range along a specific given direction is smaller by a factor of $1/\sqrt{3}$ than the value given by eq. (1), i.e. $\langle x^2 \rangle^{1/2} = \lambda/\sqrt{3}$.

Differently from the case of one-dimensional diffusion, the reciprocal mean range of two- or three-dimensionally migrating defects is given by the square root of the corresponding sink strength. For the latter case, the sink strength of defect clusters is proportional to the effective trapping radius, r_{ij}, whereas in the former case, it depends only logarithmically upon r_{ij}. For three-dimensionally migrating defects the sink strength may be written approximately as

$$k_i^2 = \lambda_i^{-2} = \sum_i 4\pi r_{ij} C_j + \rho. \tag{2}$$

A similar expression holds for defects migrating in two dimensions [39].

For applications of eqs. (1) and (2), information about effective trapping radii and cross sections are required. Since the elastic interaction of small loops with cavities is weak and of short range the trapping radii and cross sections may be approximated in this case by the corresponding geometrical expressions.

$$r_{lc} = r_l + r_c \approx r_c \text{ and } \sigma_{lc} = \pi(r_l + r_c)^2 \approx \pi r_c^2, \tag{3}$$

where r_l and r_c are the radii of loops and cavities, respectively.

For the interaction of gliding loops with sessile loops and network dislocations the situation is more complicated. In these cases, the trapping radii and cross sections are defined by the long range elastic interaction between the respective components. A fully quantitative evaluation of r_{ij} and σ_{ij} is rather involved because of the complicated spatial dependence of elastic interactions, in particular for elastically anisotropic media. Since the effective trapping radii turn out to be large compared with the geometrical radii of the loops, the "infinitesimal loop approximation" [40] may, however, be applied. Using this approximation, representative estimates for r_{ij} may be obtained by considering appropriate directional averages of the respective elastic interaction in an elastically isotropic effective medium. Since the linear directional average of the elastic interaction vanishes we use its directional standard deviation. The distance below which this quantity is smaller than the thermal energy kT may then be considered to represent an upper bound estimate for the trapping radius r_{ij}. For the interaction of two loops containing n_i and n_j SIAs or vacancies, and for the interaction of a loop i with a straight dislocation, this procedure results in temperature dependent interaction radii which may be approximated by, respectively,

$$r_{ij} \approx 0.5\left(n_i^g n_j^s \mu\Omega/kT\right)^{1/3}\Omega^{1/3}$$

$$\approx 1.5\left(n_i^g n_j^s T_m/T\right)^{1/3}\Omega^{1/3}$$

(loop/loop interaction), (4)

$$r_i \approx 0.1b\, n_i^g \mu\Omega/kT \approx 3.5b\, n_i^g T_m/T$$

(loop/dislocation interaction), (5)

where μ is the shear modulus, Ω is the atomic volume and b is the Burgers vector of the dislocation. At the right hand side of eqs. (4) and (5) $\mu\Omega/kT_m = 35$ has been used as an estimate on a homologous basis. Elastic anisotropy is expected to result in somewhat larger values for r_{ij} than given by eqs. (4) and (5). The mean ranges for one- and three-dimensional diffusion are obtained by inserting eqs. (3) to (5) into eqs. (1) and (2), respectively.

At extremely low trap densities, a glissile loop may be as likely to interact with another glissile loop as to get trapped by an immobile cluster. The corresponding mean range may be estimated by equating the mutual interaction rate with the trapping rate, similar to the diatomic nucleation model [41]. We estimate ranges of about 20 μm for negligible glide energies (< 0.1 eV) and of a few μm for glide energies of some tenths of an eV. (Note that these large spatial scales are of the order of the distances between loop patches observed in the extremely heterogeneous microstructures evolving in pure Cu after low dose n-irradiation [17]).

In figs. 2a to 2c, the mean ranges of glissile loops related to trap distances are plotted versus trap densities. For comparison, the mean ranges of two- and three-dimensionally diffusing defects are included. The most striking feature revealed by fig. 2 is that the

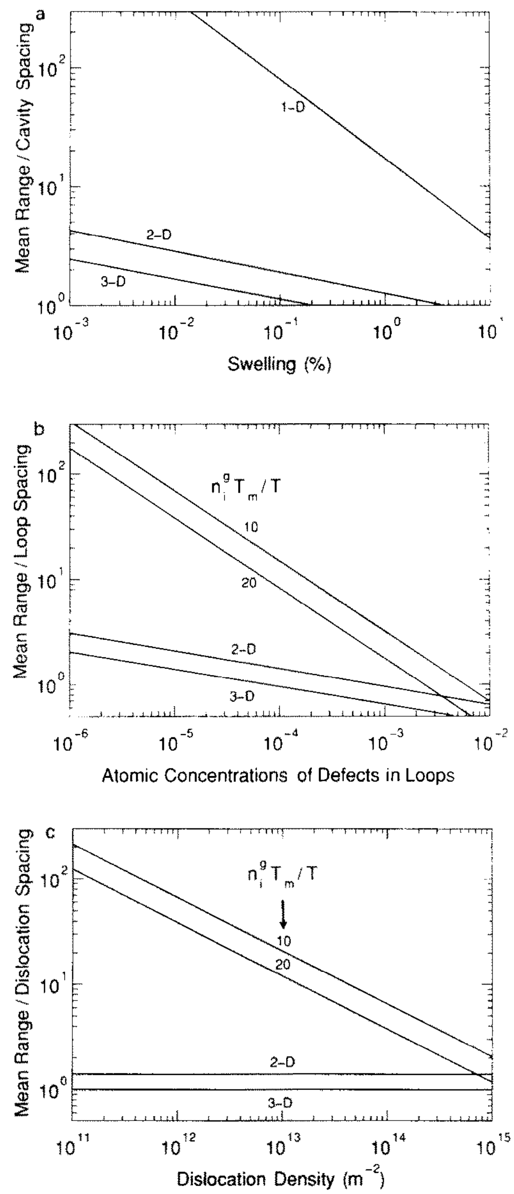

Fig. 2. Ratio of mean range and trap distances versus trap concentrations for small defect clusters migrating in one, two and three dimensions in a microstructure dominated by (a) cavities, (b) small immobile clusters, and (c) dislocations, respectively. At medium and low trap concentrations, the range for 1D migration is substantially larger than for 2D and 3D migration.

ranges for the one-dimensional loop glide are significantly larger than for 3D migration. The difference is much more substantial than between 2D and 3D migration. (In the corresponding figure of ref. [34], the mean range in a given direction has been plotted for a random distribution of the diffusion directions over equivalent directions of a cubic lattice, representing an upper bound estimate for the mean range of a one-dimensional diffusional zigzag motion.)

Fig. 2 shows that for well annealed metals with dislocation densities $< 10^{12}$ m^{-2}, grains with diameters up to 100 μm would be virtually transparent for gliding loops at the beginning of irradiation. This would remain true during the initial stages of the microstructural evolution when cavity nucleation takes place. At later stages, gliding loops produced in a region adjacent to a planar sink would be preferentially trapped by such a sink. The width of this region is given by the mean range of the loops and decreases proportionally with this but reaches the μm range only at a well developed microstructure. In section 5, these features will be used in analysing the enhanced swelling in regions adjacent to grain boundaries.

An interesting question concerns the fate of glissile loops once trapped by a sessile loop or a dislocation. Unless a glissile loop experiences a close encounter with the trap it would have to change its Burgers vector to get definitively absorbed. If such a configurational transition did not happen the glissile loop could either be detrapped by thermal activation or it could assist the trapping of another glissile loop. The binding of such a loop complex to the primary trap would be stronger than the one of a single loop and the detrapping probability would be correspondingly reduced. By such trapping sequences, a dislocation could catalyse the formation of sessile loops in its vicinity. This mechanism provides the possibility to explain the decoration of dislocations in well annealed metals irradiated at low doses [17,25].

4. Defect accumulation in the early transient

The defect reactions occurring at the beginning of irradiation can be decisive for the subsequent microstructural evolution, in particular in well annealed pure metals. Even this early stage of the microstructural evolution is rather complex under cascade damage conditions and would require, in principle, a description by an extremely complicated system of kinetic equations. In order to get a first insight into the main trends, it is sufficient, however, to consider only the most important reactions between a few representative microstructural components. Here, interest will be focused on the evolution of immobile SIA and vacancy clusters and the building up of a vacancy supersaturation by the decay of the vacancy clusters in a stage prior to void nucleation.

The single defects and defect clusters which survive intracascade recombination and are thus available for intercascade reactions are responsible for the global microstructural evolution. These defects represent only a small fraction (around 10%) of the initial displacements predicted by the NRT model. We assume that these SIAs and vacancies are produced at a total rate P consisting of fractions $(1-\epsilon_i)$ and $(1-\epsilon_v)$ of single SIAs and vacancies and of fractions and ϵ_i and ϵ_v of SIAs and vacancies in clusters containing n_i SIAs and n_v vacancies, respectively. The fraction of SIAs in clusters is subdivided into partial fractions of sessile clusters and glissile loops, $\epsilon_i = \epsilon_i^s + \epsilon_i^g$. The main microstructural component considered in addition to the primary point defects and their clusters is a dislocation network of density ρ which is assumed to be constant or to vary slowly compared to the buildup of the primary defects.

The point defect reactions considered are mutual recombination and annihilation at sinks while intercascade point defect clustering is neglected. Two types of cluster reactions are considered: (1) The biased absorption of point defects and the thermal decay of vacancy clusters. For simplicity, the effective shrinkage rates are assumed to be constant during the lifetimes of the clusters. (2) The absorption of gliding loops by sessile clusters according to their fractional sink strengths for the one-dimensional glide motion.

With these assumptions the evolution of the atomic concentrations c_i, c_v, c_{ni} and c_{nv} of SIAs, vacancies and their clusters, respectively, may be described in terms of the following set of kinetic equations

$$\partial c_i/\partial t = (1-\epsilon_i)P - aD_i c_i c_v - \bar{z} D_i c_i k^2, \tag{6a}$$

$$\partial c_v/\partial t = (1-\epsilon_v)P - aD_i c_i c_v - D_v\left(c_v - \overline{c_v^d}\right)k^2, \tag{6b}$$

$$n_i \partial c_{ni}/\partial t = \epsilon_i^s P - \left[(D_v c_v - z_{ni} D_i c_i)k_{ni}^2 - \epsilon_i^g P \kappa_{ni}/\kappa\right]_+, \tag{6c}$$

$$n_v \partial c_{nv}/\partial t = \epsilon_v^s P - \left[(z_{nv} D_i c_i - D_v c_v + D_v c_v^{nv})k_{nv}^2 + \epsilon_i^g P \kappa_{nv}/\kappa\right]_+, \tag{6d}$$

with

$$k^2 = \rho + k_{ni}^2 + k_{nv}^2, \quad k_{ni,nv}^2 = a_{ni,nv} c_{ni,nv},$$

$$\kappa_{ni,nv} = \sigma_{ni,nv} c_{ni,nv}/\Omega.$$

Here, a is the recombination coefficient; $D_i \gg D_v$ are the SIA and vacancy diffusivities, respectively; k^2 is the total sink strength for the annihilation of vacancies, k_{ni}^2 and k_{nv}^2 are the partial sink strengths of SIA and vacancy clusters, respectively; z denotes bias factors accounting for the preferential absorption of SIAs, $\bar{z}$ is the corresponding average over all sinks; κ is the total sink strength for the annihilation of glissile loops at defect clusters and dislocations and $\kappa_{ni,nv}$ are the partial sink strengths of SIA and vacancy clusters which are generally small (note that k and κ represent reciprocal mean defect ranges for 3D and 1D migration, respectively); c_v^d is the local equilibrium vacancy concentration at the boundary of defect d (with $d = n$v, ni, ρ), and $\overline{c_v^d}$ is its average over all defects; a_{ni} and a_{nv} are the trapping coefficients of the SIA and vacancy clusters, σ_{ni} and σ_{nv} are their effective cross sections for trapping glissile loops. The suffix + at the right hand side of the square brackets in eqs. (3c) and (3d) indicates that the corresponding term is non-zero only if the argument is positive, $[x]_+ = (x + |x|)/2$. Note that eqs. (3c) and (3d) would describe the temporal change of the SIA and vacancy concentrations deposited in clusters, $\partial n_{i,v} c_{ni,nv}/\partial t$, if the square bracket term were not restricted in this way. Eqs. (6a) to (6d) represent an extension of a previously suggested set of equations [43].

Inspite of the large number of parameters involved in eqs. (6a)–(6d) (see table 1) some general trends can be deduced. Fortunately, it turns out that some of the general features do not crucially depend upon the details of the chosen parameters. For the effects resulting from the spontaneous production of mobile and immobile defect clusters, for instance, it is only important that the corresponding fractions considered are sufficiently large. Activation energies characterizing defect properties, on the other hand, generally scale with the melting temperature of the metal. This makes it possible to discuss the results in terms of a homologous temperature T/T_m [44].

Some of the parameters of eq. (6) deserve more detailed explanation. In cases where the concentration of immobile SIA loops is limited by the absorption of excess vacancies, shrinking SIA loops would eventually reach a size at which they become glissile. In such cases, a lower bound value of the fraction of SIAs in glissile loops is given by the fraction of sessile loops, ϵ_i^s, times the ratio of the size when the loops become glissile over the average size at the time of production.

The bias factors z for the preferential absorption of SIA by defect clusters and dislocations are all assumed to be equal, for simplicity. As long as these types of sinks are dominant z may be considered to be a part of the effective SIA diffusivity. An explicit reference to this parameter is only required when discussing its effect on void nucleation and growth.

The trapping coefficients $a_{ni,nv}$ of large loops (and voids) for 3D diffusion increase linearly with the radius of these clusters. The trapping efficiency of small loops is, however, underestimated by the corresponding expressions. For small loops produced in cascades, we have therefore chosen a value of 5×10^{20} m^{-2} (as for the recombination coefficient) which is consistent with eq. (6) for the trapping of single SIAs by loops containing 25 point defects at about 0.4 T_m [32].

The driving force for the thermal decay of large vacancy loops may be described in terms of the dislocation line tension. In using this continuum approach, we choose the effective dislocation core radius such that the energy of small vacancy clusters consisting of only a few vacancies, may be equally described as loops or voids. There is experimental and theoretical evidence that the local equilibrium vacancy concentration and the decay rate of small vacancy loops is overestimated by such an approach [45]. In assuming, however, that the local vacancy concentration and the decay rate remain constant at their initial levels during the decay process, this overestimation is, at least partly, compensated. In this approximation, the lifetime resulting from thermal vacancy emission may be written as

$$n_v(\tau_{nv}^{th})^{-1} = D_v(c_v^{nv} - c_v^{eq})a_{nv} = D^{sd}(c_v^{nv}/c_v^{eq} - 1)a_{nv}$$
$$\equiv (D^{nv} - D^{sd})a_{nv}, \quad (7)$$

where $D^{sd} \approx D_v c_v^{eq}$ is the equilibrium self-diffusion coefficient, c_v^{eq} is the global equilibrium vacancy concentration in the absence of radiation damage and D^{nv} would define a self-diffusion coefficient in the presence of an inexhaustable concentration of vacancy clusters.

A full understanding of the microstructural evolution requires an analysis of its transient behaviour. The buildup period of the point defects and their clusters may be divided into the following characteristic stages:

(1) Buildup of the cascade induced defects without any significant intercascade reaction until a substantial fraction of the SIAs produced have annihilated at sinks. At the end of this stage and beyond, the SIA concentration is in quasi-steady-state and decreases with increasing sink strength. At swelling temperatures, this reactionless stage is extremely short ($< 10^{-10}$ dpa) for normal dislocation densities $\rho > 10^{10}$ m^{-2} because of the high mobility of the SIAs.

(2) Increase in the vacancy concentration until it has reached quasi-steady-state. For the conditions considered here, this occurs around 10^{-5} NRT dpa.

(3) Increase in the SIA and vacancy cluster concentrations up to intermediate quasi-steady-state levels. These levels are controlled by the absorption of excess vacancies resulting from loop escape and the thermal emission of vacancies, respectively. The duration of this stage is essentially defined by the thermal lifetime of the vacancy clusters and is longer at low swelling temperatures ($\leq 0.35\ T_m$) and shorter at high swelling temperatures ($\geq 0.35\ T_m$) than the time needed for vacancies to reach a quasi-steady-state.

(4) Increase in the concentration of one of the cluster components at a reduced rate, depending on the dislocation density, due to the unbalance between the vacancy flux to dislocations and the SIA flux associated with SIA loop glide until the final quasi-steady-state of the cluster densities is reached, as discussed in the preceding subsection. In most cases this is completed below 10^{-2} NRT dpa.

(5) Increase in the densities of secondary microstructural components such as dislocations and cavities (not covered by the present approach) until the whole microstructure reaches a quasi-steady-state.

We have calculated the evolution of immobile SIA and vacancy clusters and the building up of a vacancy supersaturation assuming that glissile loops are predominantly absorbed by extended sinks such as dislocation and grain boundaries, $\kappa_{ni,nv}/\kappa \to 0$. This assumption is expected to be justified at the beginning of

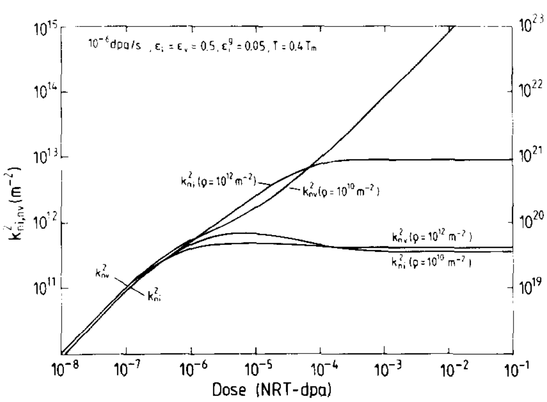

Fig. 3. Sink strengths and number densities of SIA and vacancy clusters versus NRT dose (= 10 Pt) at 0.4 T_m for two values of the dislocation density ρ. Parameters not given in the figure may be found in table 1. Note the change from vacancy cluster dominance to SIA cluster dominance when the dislocation density is increased.

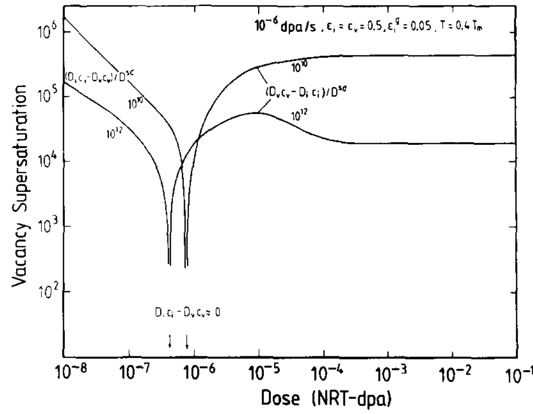

Fig. 4. Vacancy supersaturation, $|D_i c_i - D_v c_v|/D^{sd}$, according to fig. 3. Note the maximum around 10^{-5} NRT dpa for the higher dislocation density.

irradiation when the density of clusters is still low. A more detailed analysis will be given elsewhere [42].

An example is given in fig. 3, where the sink strengths and number densities of SIA and vacancy clusters developing at 0.4 T_m are plotted versus the NRT displacement dose for two values of the dislocation density, ρ, assuming the parameter values given in table 1. At about 10^{-6} NRT dpa, an intermediate quasi-steady-state controlled by the thermal decay of the vacancy clusters is reached. The most striking feature in the subsequent cluster evolution where the total sink strength of both types of clusters becomes substantially larger than ρ is the change-over from vacancy cluster dominance at the lower value of ρ to SIA cluster dominance at the higher value of ρ. Similar features are found for other temperatures, although for other values of ρ. In the more general case, the changeover occurs at a critical dislocation density around

$$\rho_c \approx \epsilon_i^g(1 + \epsilon_v^s/\epsilon_i^s)/D^{nv}. \qquad (8)$$

The high sensitivity of the cluster densities to ρ suggest the possibility of spatial and temporal instabilities in the microstructure around $\rho = \rho_c$.

The results for the evolution of the sink strengths of SIA and vacancy clusters shown in fig. 3 may be used to derive the temporal change of point defect diffusivities and fluxes such as the SIA and vacancy fluxes, $D_{i,v} c_{i,v}$, or the effective vacancy supersaturation $(D_v c_v - D_i c_i)/D^{sd}$, during the transient. In fig. 4 the latter quantity is plotted versus the NRT dose for the conditions assumed in fig. 3. $D_i c_i$ decreases from a high early quasi-steady-state value to a low late quasi-steady-state value while $D_v c_v$ shows the opposite behaviour. $D_v c_v - D_i c_i$ becomes positive when the vacancies reach quasi-steady-state, increases continuously to a late quasi-steady-state value for low ρ but passes a maximum for higher dislocation densities.

This transient behaviour deserves some attention. Microvoids that may not be stable in steady state at higher doses could nucleate during a transient maximum of the net vacancy flux (see fig. 4 for $\rho = 10^{12}$ m^{-2}) even if they would not nucleate in the late steady state. This will occur just when the point defects have reached quasi-steady-state in cases where the SIA loop number density is still increasing. It is worth noting that the levels of vacancy supersaturation are extremely high.

For most of the conditions covered by fig. 3, grains with diameters up to 100 μm would be virtually transparent for glissile loops (see fig. 2) justifying our assumption of complete escape of those loops to grain boundaries. An exception is the vacancy cluster sink strength at higher doses for the lower dislocation density. Under these conditions, the vacancy cluster density will indeed be limited by the absorption of glissile SIA loops [42].

5. Enhanced swelling adjacent to grain boundaries

The probability of a glissile loop to get trapped within a grain increases with increasing sink strength. Glissile loops produced in a region adjacent to a planar sink will, however, still be preferentially trapped by such a sink.

Since the enhanced swelling adjacent to grain boundaries has been investigated experimentally in some detail, it is most suited to test the idea of 1D SIA transport by loop glide. In Al neutron irradiated at 120°C, for instance [26], voids at low doses were first observed within a band, 5 to 10 μm wide, adjacent to the void denuded zone along grain boundaries. At doses where voids become observable throughout the grain interior a pronounced grain boundary related zone of enhanced swelling develops extending up to 15 μm into the grain interior. With increasing dose, the maximum in swelling shifts closer towards the grain boundary denuded zone and the peak becomes narrower. Similar effects have been observed in other fcc metals such as Ni [20,28] and Cu [46] as well as in the bcc metals V and Nb [22].

Enhanced swelling adjacent to grain boundaries has been discussed recently in terms of 2D SIA transport [28,39]. This reduction of the migration dimension from three or two is, however, not at all sufficient to explain the observed wide range of the swelling zone. In the following we make an attempt to model this phenomenon in terms of one-dimensional SIA transport by loop glide.

In the presence of traps, the length of the glide cylinder of a glissile loop will be limited. Accordingly, the loop flux to a specific trap is given by one half of the loop production within all glide cylinders limited by that trap and by its next nearest neighbour traps. Thus, the arrival rate of SIAs transported in glissile loops at a trap of cross section σ may be written as

$$\mathrm{d}n_i/\mathrm{d}t = (\epsilon_i^g P/\Omega)\sigma\bar{\lambda}, \tag{9}$$

where Ω is the atomic volume and $\bar{\lambda}$ is the mean free path representing the average loop supply cylinder length. For a random but generally space dependent trap distribution in an infinite space, the average loop supply cylinder length of a trap at $\boldsymbol{r}$ may be written as [35]

$$\bar{\lambda}(\boldsymbol{r}) = 0.5\int_{-\infty}^{\infty} \exp\left[-\int_0^z \kappa(\zeta)\,\mathrm{d}\zeta\right]\kappa(z)z\,\mathrm{d}z, \tag{10}$$

where κ is defined by eq. (1) and z is the coordinate in the glide direction measured from $\boldsymbol{r}$. For a homogeneous trap distribution (κ constant), integration of eq. (10) yields $\bar{\lambda} = \kappa^{-1}$ which becomes $(\sigma C)^{-1}$ for a monodisperse trap distribution of number density C (see eq. (1)). Using this relation in eq. (9) we confirm the simple partitioning relation $\mathrm{d}n_i/\mathrm{d}t = \epsilon_i^g P/(C\Omega)$.

For a trap in the vicinity of a planar sink, half of the loop supply cylinders, namely the ones in the direction of the planar sink, will be shortened by this sink. The integrals in eq. (10) must be correspondingly limited in this case. For simplicity we consider a homogeneous trap distribution in a half-infinite space limited by the planar sink. In this case, the integral expression corresponding to eq. (10) for the average loop supply cylinder length of a trap at a distance x from the planar sink yields

$$\bar{\lambda}(x, \theta) = \kappa^{-1}[1 - 0.5(1 + \kappa x/\cos\theta) \times \exp(-\kappa x/\cos\theta)], \tag{11}$$

where θ is the angle between the glide direction and the normal to the planar sink. This expression is to be averaged over all crystallographically equivalent glide directions. Approximating this discrete directional average by a continuous isotropic average we find

$$\kappa\bar{\lambda}(x) = 1 - 0.5\exp(-\kappa x). \tag{12}$$

The effect of the planar sink is described by the exponential term at the right hand side of eq. (12).

Using eq. (12) we obtain the local rate at which SIAs are lost to the planar sink, L_i, by multiplying eq. (9) with the local trap number densities, summing over

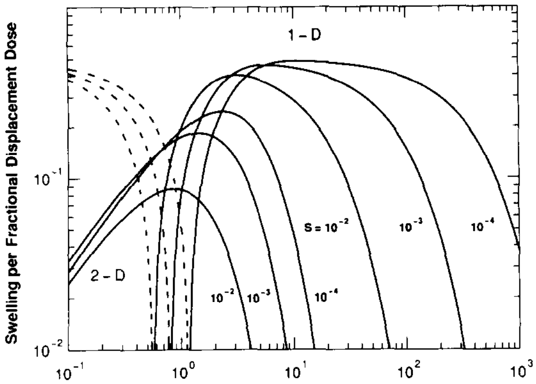

Fig. 5. Swelling per fractional displacement dose (displacements per atom resulting in low-dimensional SIA transport) versus distance from a planar sink normalized to cavity spacing for 1D and 2D SIA transport assuming three different swelling levels. At small distances, the swelling rates for 1D SIA transport formally become negative as indicated by broken lines.

all trap species and taking the difference between the resulting expressions for the bulk ($x \to \infty$) and the zone adjacent to the planar sink (finite x),

$$L_i = 0.5\epsilon_i^g P \exp(-\kappa x). \tag{13}$$

The local loss rate of the excess vacancies corresponding to the SIAs escaping in the form of loops is obtained by solving the diffusion equation for this one-dimensional problem [28]

$$L_v = \epsilon_i^g P \exp(-kx), \tag{14}$$

where k^2 is the local sink strength for the absorption of vacancies. When voids form the dominant sinks (in addition to the planar sink), the swelling rate, dS/dt, is simply given by the difference $L_i - L_v$, i.e.

$$dS/dt = \epsilon_i^g P[0.5 \exp(-\kappa x) - \exp(-kx)]. \tag{15}$$

An analogous expression (but without a factor of 0.5 in the SIA contribution) holds for SIA loss by 2D SIA transport [28,39]. Note that at small x, where the vacancy loss is in excess of the SIA loss, dS/dt is negative indicating a stabilization of an existing void denuded zone. In this region, an existing void would absorb more SIAs than vacancies and thus shrink away.

In fig. 5, the swelling rate is plotted versus distance from the planar sink for 1D as well as 2D interstitial transport assuming three different swelling levels. The plots show that the range as well as the magnitude of the effect are significantly higher in the 1D than in the 2D case. The broken lines below 1 cavity spacing indicate negative swelling rates. The width of the peak swelling zone is found to be approximately equal to the

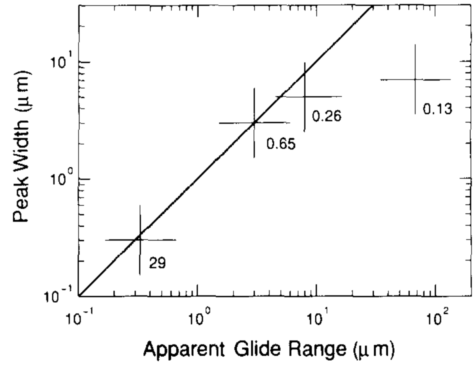

Fig. 6. Width of the swelling peak (full width at half maximum) versus apparent loop glide range corresponding to maximum swelling for Al n-irradiated to doses between 0.13 and 29 NRT dpa. The proportionality and equality of both length scales at the two higher doses indicate that voids form the dominant traps for SIA loops in the peak swelling zone in this dose range.

mean range corresponding to peak swelling. Both length scales are decreasing with increasing swelling.

This result may now be compared with experimental observations. In fig. 6, the width of the peak swelling zone is plotted against the apparent SIA loop glide range corresponding to the maximum swelling for neutron-irradiated Al. The approximate proportionality and equality of both length scales at high doses indicates that voids form the dominant traps for SIA loops gliding through the peak swelling zone in this dose range. At low doses, the mean range is obviously reduced by other defects, most likely by small sessile dislocation loops.

A detailed calculation of the spatial and temporal evolution of the swelling adjacent to grain boundaries is more complicated. An approximation for the maxi-

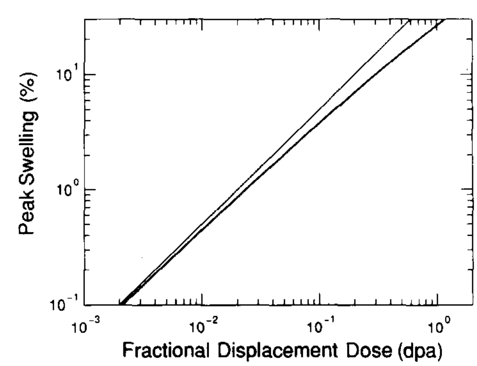

Fig. 7. Peak swelling versus fractional displacement dose, $\epsilon_i^g Pt$ (fat line). The thin line indicates maximum vacancy accumulation corresponding to complete SIA loop escape.

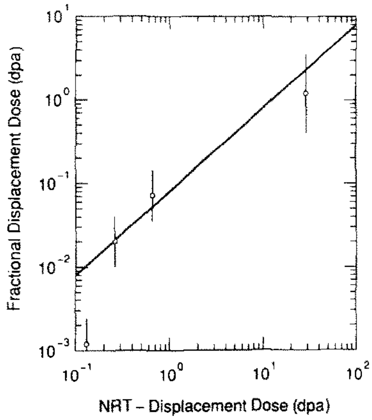

Fig. 8. Fractional displacement dose, $\epsilon_i^g\ Pt$, derived from observed peak swelling versus NRT displacement dose according to fig. 8. The straight line through the high dose values corresponds to 4% NRT displacements assumed to result in glissile SIA loops.

mum swelling is obtained by integrating the maximum swelling rate following from eq. (15). The result is shown in fig. 7. At low doses, the vacancy accumulation in the peak swelling zone corresponds to the escape of half of the SIA loops. At higher doses, swelling is somewhat reduced due to increased SIA loop capture by existing voids.

For cases where voids are the dominant sinks, fig. 7 may now be used to deduce the fractional displacement dose, $\epsilon_i^g Pt$ (displacements per atom resulting in glissile loops), from observed peak swelling data. For given dose, this allows us to estimate the fraction of displacements resulting in glissile loops. In fig. 8, the fractional displacement dose, $\epsilon_i^g Pt$, is plotted versus the NRT dose for neutron-irradiated Al. The straight line through the high dose values, for which voids have been shown to form the dominant traps for glissile SIA loops in the peak swelling zone, corresponds to 8% of the NRT displacements assumed to result in glissile loops. The experimental uncertainties and the relative crudeness of the approximations allow, however, a wide range of values between, say, 5 and 12%. A preliminary analysis of swelling in cells of polygonized Al confirms this estimate. Inspite of the uncertainties, we may conclude that in Al a substantial fraction of the SIAs surviving intracascade recombination appear to be transported in the form of loops.

At first, the observation of enhanced swelling in the vicinity of grain boundaries in Al bombarded with 600 MeV protons at 393 K [24,25] may seem to be at variance with the above analysis. In the latter case, the peak zone is significantly narrower and closer to the grain boundary than in the neutron case. Even at the highest dose examined (2 NRT dpa) the 1D mean range through the observable void microstructure in the peak zone is about 6 μm whereas the zone width at half of the maximum excess swelling is only about 2 μm. Irradiation experiments at higher temperature and higher dose (403 and 413 K, 5 NRT dpa [27]) reveal the origin of this apparent discrepancy. At these conditions small bubbles become visible which have obviously controlled the width of the peak zone.

The evidence for 1D SIA transport provided by the above analysis does not necessarily mean evidence for loop glide. Any other type of 1D transport of atoms from the grain interior into grain boundaries would result in similar effects if the range of such a transport was sufficiently large [28]. Mechanisms such as replacement sequences, channeling and crowd ion migration are, however, unlikely to account for SIA transport over the required distances of several μm.

6. Conclusions

Diffuse X-ray scattering experiments as well as MD simulation studies provide evidence for substantial SIA clustering in multidisplacement cascades. In addition, both types of studies indicate that small SIA clusters are mobile. The highly heterogeneous microstructures evolving in well annealed pure metals during low dose irradiation provide indirect evidence for long-ranging SIA transport, most likely by loops formed in cascades.

Small glissile loops are expected to perform an extremely fast diffusive glide motion. Except for very high defect densities the range of one-dimensionally gliding loops is substantially larger than that of three-dimensionally migrating defects. By glide, SIA loops may reach extended sinks such as dislocations, or grain boundaries. This SIA cluster removal mechanism represents an efficient driving force for void swelling.

The defect accumulation in the early transient may be described by a simple set of kinetic equations in which the escape of glissile SIA loops is taken into account but reactions between clusters are neglected. Above 0.35 of the melting temperature and for low to moderate dislocation densities, the number densities of immobile vacancy and SIA clusters are most likely to be limited by the thermal decay and by the absorption of the excess vacancies associated with the loss of glissile SIA loops to extended sinks, respectively. At a critical dislocation density, the microstructural dominance changes from vacancy clusters to SIA clusters. This high sensitivity of the cluster densities to the dislocation density suggests the possibility of spatial and temporal instabilities in microstructure.

The effective vacancy supersaturation resulting from the SIA loss by loop escape is generally found to be very high. At sufficiently high dislocation densities, the vacancy supersaturation passes a maximum during the transient. Microvoids that are not stable in the quasi-steady-state at higher doses could be stabilized during the transient maximum of the vacancy supersaturation.

The enhanced swelling observed adjacent to grain boundaries may be modeled in terms of SIA loop escape to these planar sinks. The width of the swelling zone turns out to be of the order of the mean range of loop escape. From the maximum swelling rate observed in neutron-irradiated Al, the fraction of SIAs produced in the form of glissile loops is estimated to be between 5 and 12%. This certainly represents a substantial fraction of the SIAs surviving intracascade recombination. More accurate experimental data and more detailed modeling are required to get more precise numbers, in particular for other metals.

References

[1] W. Jäger, J. Micros. Spectros. Electron. 6 (1981) 437.
[2] C.A. English and M.J. Jenkins, Mater. Sci. Forum 15–18 (1987) 1003.
[3] M. Kiritani, ibid., p. 1023.
[4] B. von Guerard, D. Grasse and J. Peisl, Phys. Rev. Lett. 44 (1980) 262.
[5] D. Grasse, B. von Guerard and J. Peisl, J. Nucl. Mater. 120 (1984) 304.
[6] R. Rauch, J. Peisl, A. Schmalzbauer and G. Wallner, J. Nucl. Mater. 168 (1989) 101.
[7] T. Diaz de la Rubia and M.W. Guinan, J. Nucl. Mater. 174 (1990)151; Phys. Rev. Lett. 66 (1991) 2766.
[8] A.J.E. Foreman, C.A. English and W.J. Phythian, Phil. Mag. A 66 (1992) 655 and 671.
[9] R.W. Balluffi, J. Nucl. Mater. 69&70 (1978) 240.
[10] H.R. Schober and R. Zeller, ibid., p. 341.
[11] O. Bender and P. Ehrhart, J. Phys. F 13 (1983) 911.
[12] P. Ehrhart and R.S. Averback, Philos. Mag. A 60 (1989) 283.
[13] B.C. Larson and F.W. Young, Phys. Status solidi A 104 (1987) 27.
[14] R. Rauch, J. Peisl, A. Schmalzbauer and G. Wallner, J. Phys. Condens. Matter 2 (1990) 9009.
[15] W. Jäger, W. Frank and K. Urban, Radiat. Eff. 46 (1980) 47.
[16] K. Kiritani, Ultramicroscopy 39 (1991) 135.
[17] B.N. Singh, T. Leffers and A. Horsewell, Philos. Mag. A 53 (1986) 233.
[18] C.A. English, B.L. Eyre and J.W. Muncie, Harwell Report, AERE-R-12188, September 1986.
[19] S.J. Zinkle and B.N. Singh, to be published.
[20] J.O. Stiegler and E.E. Bloom, Radiat. Eff. 8 (1971) 33.
[21] K. Farrell, J.T. Houston, A. Wolfenden, R.T. King and A. Jostsons, Radiation-Induced Voids in Metals (Albany, New York, 1971) p. 376.
[22] F.W. Wiffen, Proc. Int. Conf. on as ref. [21], p. 386.
[23] M.V. Green, S.L. Green, B.N. Singh and T. Leffers, J. Nucl. Mater. 104 (1981) 1221.
[24] B.N. Singh, T. Leffers, W.V. Green and S.L. Green, J. Nucl. Mater 105 (1982) 1.
[25] B.N. Singh, J.B. Bilde-Sørensen, T. Leffers, L.S. Ozhigov and V.V. Gann, J. Nucl. Mater. 122–123 (1984) 542.
[26] A. Horsewell and B.N. Singh, ASTM STP 955 (1988) 220.
[27] M. Victoria, W.V. Green, B.N. Singh and T. Leffers, ibid., p. 233.
[28] A.J.E. Foreman, B.N. Singh and A. Horsewell, Mater. Sci. Forum. 15–18 (1987) 895.
[29] C.H. Woo and B.N. Singh, Phys. Status Solidi B 159 (1990) 609.
[30] C.H. Woo and B.N. Singh, Philos. Mag. A 65 (1992) 889.
[31] B.N. Singh, C.H. Woo and A.J.E. Foreman, Mater. Sci. Forum 97–99 (1992) 75.
[32] B.N. Singh and A.J.E. Foreman, Philos. Mag. A 66 (1992) 975.
[33] C.H. Woo, A.A. Semenov and B.N. Singh, in these Proceedings (Workshop on Time Dependence of Radiation Damage Accumulation, Montreux, Switzerland, 1993), J. Nucl. Mater. 206 (1993) 170.
[34] H. Trinkaus, B.N. Singh and A.J.E. Foreman, J. Nucl. Mater 199 (1992) 1.
[35] H. Trinkaus, unpublished work.
[36] W.M. Lomer and A.H. Cottrell, Philos. Mag. 46 (1955) 711.
[37] U. Gösele and A. Seeger, Philos. Mag. 34 (1976) 177.
[38] K. Schroeder, Radiat. Eff. 5 (1970) 255;
W. Schilling, K. Schroeder and H. Wollenberger, Phys. Status Solidi 38 (1970) 245.
[39] J.H. Evans and A.J.E. Foreman, J. Nucl. Mater. 137 (1985) 1.
[40] F. Kroupa, Philos. Mag. 7 (1962) 783.
[41] H. Trinkaus, J. Nucl. Mater. 133&134 (1985) 105; Radiat. Eff. 101 (1986) 91.
[42] H. Trinkaus, B.N. Singh and A.J.E. Foreman, to be published.
[43] H. Trinkaus, B.N. Singh and A.J.E. Foreman, J. Nucl. Mater. 174 (1990) 80.
[44] H. Trinkaus, ibid., p. 178.
[45] A.J.E. Foreman and M.J. Makin, J. Nucl. Mater. 79 (1979) 43.
[46] B.N. Singh and A. Horsewell, ICFRM6, to be published in J. Nucl. Mater.